Flip-Consistent Training: A Plug-In Loss for Generalizable Polyp Segmentation

Authors

***Abstract*—Polyp segmentation networks lose accuracy on centres unseen in training, and their predictions change when an image is mirrored, although a mirrored polyp is still a polyp. We propose flip-consistent training (FCT), a plug-in training loss that leaves the network and its inference unchanged. FCT supervises each image in three orientation views, upright and flipped horizontally and vertically, and adds a consistency term that draws the predictions of the views together. We show that, for binary labels, such a term acts on each pixel as the supervised loss with an adaptive weight, which is negative where a view is drawn toward a less accurate one. FCT therefore weights its consistency term with the label, so that no view is drawn toward a misclassification it has avoided, with a single constant set by the decision threshold. Experiments with three architectures on five public benchmarks, each repeated over six random seeds, show that FCT improves Dice on the unseen ETIS dataset for all three networks, and the five-set mean on every seed for Polyp-PVT and PraNet-V2, whose flip inconsistency it also markedly reduces. A study of the loss terms attributes most of the ETIS gain to the multi-view supervision; the full loss, with its consistency term and weights, is the only variant that improves all five test sets. Notably, FCT improves PraNet-V2 on ETIS by 3.6 Dice points.**

***Index Terms*—Polyp segmentation, domain generalization, training loss, consistency regularization, flip equivariance.**

I. Introduction

Polyp segmentation networks that are accurate on their training data lose much of that accuracy at new centres, whose images differ in endoscopes, illumination and patients [1], [2]. This gap limits their clinical value: colorectal cancer is among the leading causes of cancer death [3], and automatic segmentation aims to help endoscopists, who miss about one in five polyps [4]. Recent networks reach high Dice on test images from their training datasets [5]–[9], but much lower Dice elsewhere. In our experiments, for example, Polyp-PVT [7] reaches 93.2% mean Dice on CVC-ClinicDB but only 78.6% on ETIS, a dataset from another hospital. For clinical use, closing this gap matters more than further gains on the training distribution.

Polyp segmentation also has a symmetry that standard training does not enforce. A mirrored polyp is still a polyp, so the correct mask of a mirrored image is the mirrored mask. We measure departures from this flip equivariance with the flip gap: one minus the Dice overlap between the binary masks predicted for an image and for its flipped copy, once flipped back, averaged over horizontal and vertical flips. As Fig. 1 shows, the flip gap of trained models is far from zero, and it is largest on the two hardest test sets, CVC-ColonDB and ETIS. We hypothesize that a model whose answer changes with a mirror image relies partly on cues that do not belong to the polyp, and that such cues are unlikely to transfer to new centres.

Two remedies are common, and both act on the training loss. Flip augmentation trains on mirrored copies with mirrored labels, but each copy is constrained only by its own label, never by the prediction on its mirror. Consistency regularization adds that constraint with a loss term that penalizes disagreement between the predictions on the views. It comes from semi-supervised learning, where agreement between views is the only signal that unlabeled images provide [10]–[12], and medical variants enforce it under spatial transformations [13], [14]. Since no label says which view is right, the views are treated as equally reliable or weighed by the model's own confidence. Supervised uses of consistency, such as AugMix [15] and R-Drop [16], keep this symmetry even though the label is known, so a correct view can be drawn toward a wrong one.

We show that, under full supervision, a consistency term is a reweighted supervised loss. For binary segmentation, the gradient of a cross-entropy consistency term that draws one view toward the detached prediction of another equals the supervised gradient, scaled by one minus the ratio of the two views' errors. Similar decompositions of the distillation gradient are known for classification [17], [18]; here we apply the identity pixel by pixel, across views of one image. A consistency term is therefore an adaptive pixel weight on the supervised loss: positive where a view is farther from the label than its target, and negative where it is closer. Unweighted consistency mixes both kinds of pixels in every image, and only the label can tell them apart.

Building on this identity, we propose flip-consistent training (FCT), a plug-in training loss for polyp segmentation. FCT supervises each image in three orientation views, upright, flipped horizontally and flipped vertically, and adds a consistency term that draws the predictions of the views together: each flipped view toward the upright view, and the upright view toward the flipped view closest to the label. Label-aware weights complete the loss. A flipped view moves toward the upright view at full strength where it is farther from the label; where it is closer, it moves only if the upright view is correct, in proportion to how far the upright prediction lies on the correct side of the 0.5 threshold. This weak term keeps the views consistent without moving either across the threshold, much as label smoothing tempers overconfident targets [19], and its single constant is set by the decision threshold rather than tuned. FCT changes only the training loss: it multiplies the training time by about 3.5 and adds nothing at inference. To the best of our knowledge, FCT is the first consistency loss that uses the label to set, pixel by pixel, the direction in which the views of an image are drawn together.

We add FCT, without retuning, to three segmentation networks of different design: Polyp-PVT [7], built on a pyramid

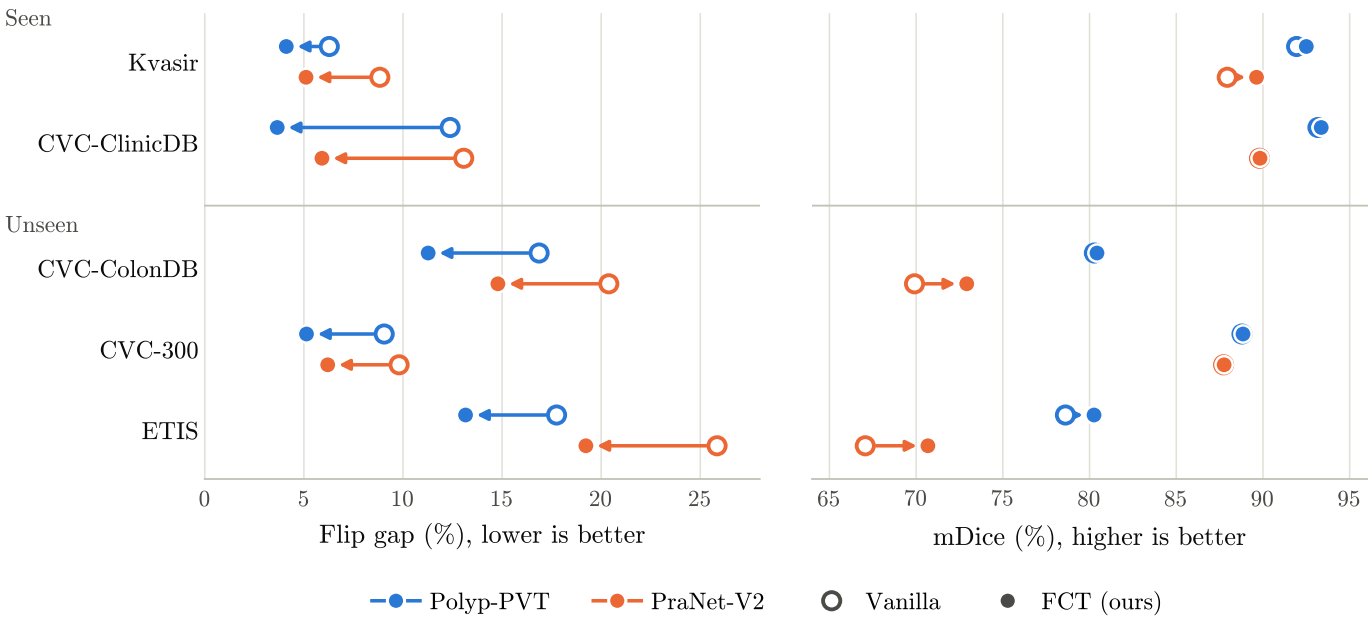


Fig. 1. Flip gap (left; one minus the Dice overlap between the binary masks predicted for an image and for its flipped copy, once flipped back, averaged over horizontal and vertical flips) and mDice (right) on five test sets, without (hollow) and with (filled) FCT. Both networks are least flip-consistent on CVC-ColonDB and ETIS, the two sets where they are least accurate. FCT lowers the flip gap on every set, and its largest mDice gain, for both networks, is on ETIS. Means over six seeds.

vision transformer; PraNet-V2 [8], a convolutional network with dual-supervised reverse attention; and ASGNet [9], which guides its features with spectral cues and already trains with random flips. Each network is trained with and without FCT under six paired random seeds and tested on five public test sets, three of which are unseen in training. FCT improves ETIS for all three networks, by +1.65, +3.61 and +1.69 points, and CVC-ColonDB for PraNet-V2, by +3.01 points, all with paired 95% confidence intervals above zero. It raises the five-set mean Dice on every seed for Polyp-PVT and PraNet-V2, and on five of six for ASGNet, whose gain is borderline. FCT also lowers the five-set mean flip gap of Polyp-PVT and PraNet-V2 by 40% and 34%, respectively. On six more centres from the PolypGen benchmark [1], a label-aware variant improves the pooled Dice on every seed (FCT itself is pending). A study of the loss terms on Polyp-PVT shows that the multi-view supervision accounts for most of the gain on ETIS, and that the consistency term and its weights complete it: FCT is the only variant that improves all five test sets on average (one row of the study is still running).

Our contributions are summarized as follows:

- We propose FCT, a plug-in training loss that supervises three orientation views of each image and draws their predictions together with a label-weighted consistency term. It changes no architecture and adds no cost at inference.
- We show that, for binary labels, a consistency term between two views acts pixel by pixel as the supervised loss with an adaptive weight, which is negative where a view is drawn toward a less accurate one, and we derive label-aware weights whose single constant is set by the decision threshold.
- We evaluate FCT on three architectures, five test sets and six paired seeds with confidence intervals, and dissect the loss into its terms. It improves ETIS for all three networks, raises the five-set mean Dice on every seed for two of them, and makes the predictions of both markedly more flip-consistent.

II. Related Work

A. Polyp Segmentation and Generalization

Deep polyp segmentation has advanced mainly through network design. PraNet [5] aggregates high-level features with a parallel partial decoder and refines boundaries with reverse attention. SANet [6] adds shallow attention and colour exchange, and UACANet [20] attends to the uncertain region of each prediction. Later networks adopt transformer encoders, as Polyp-PVT [7] and ASGNet [9] do, the latter guiding its features with spectral information, or add supervision, as PraNet-V2 [8] does with dual-supervised reverse attention. Other recent designs add contrastive transformer modules [21], boundary-sensitive attention [22] or uncertainty modeling [23], or adapt the Segment Anything Model with a convolutional branch [24]. Closest to our motivation, VANet [25] observes that steering the colonoscope tip changes the viewpoint and the appearance of polyps, and handles this with viewpoint-aware modules. These networks change the architecture, whereas FCT changes only the training loss, so it complements rather than replaces them.

Most of these models follow the protocol of [5]: they train on Kvasir-SEG [26] and CVC-ClinicDB [27], and they are tested on these two datasets and on CVC-ColonDB [28], CVC-300 [29] and ETIS [30], which are unseen in training. Accuracy on the unseen sets stays well below that on the

seen ones, and multi-centre benchmarks confirm the gap at scale [1], [2]. Domain generalization offers training-side remedies [31], such as shape-aware meta-learning [32] and frequency-space interpolation across sites [33]. For polyps, randomized global illumination augmentation [34] and colour transfer [23] target nuisance factors that vary across centres, and domain adaptation learns from unlabeled images of the target domain [35]. These methods diversify the training data, learn from differences between domains or need target images, whereas FCT adds a constraint between views of each training image and needs no target data, so the approaches are complementary.

B. Consistency Regularization and Equivariance

Consistency regularization asks a network to agree with itself across perturbed views of an input. It was developed for semi-supervised learning, where unlabeled images provide only this signal: the Π-model and temporal ensembling [10], mean teachers [11], unsupervised data augmentation [36] and FixMatch [12]. FixMatch keeps only pseudo-labels whose confidence exceeds a threshold, a proxy for correctness when the true label is unknown. Semi-supervised segmentation methods perturb inputs, features or networks [37]–[39]. In medical imaging, consistency under spatial transformations exploits the equivariance of segmentation on unlabeled scans [13], [14], or its invariance to the surrounding context [40]. Mean teachers restrict consistency to reliable regions [41] or to ambiguous ones [42], bidirectional patch displacement builds new samples for consistency learning guided by prediction confidence [43], and semi-supervised polyp segmentation aligns the affinity maps of a student and a teacher [44].

With labels available, consistency also serves as a regularizer: AugMix [15] matches predictions across augmentation chains, and R-Drop [16] across dropout masks. A sample-efficiency analysis shows that, with label-invariant augmentations, consistency regularization learns from fewer samples than training on the augmented copies with their labels [45]. Flip and rotation augmentation, by contrast, supervises each transformed copy with its own transformed label and does not relate the predictions on different copies. Equivariance can also be built into the architecture [46], [47], enforced on class activation maps in weakly supervised segmentation [48], or approximated at test time by averaging predictions over transformed inputs [49], at the price of a constrained architecture or a multiplied inference cost. All these methods treat the views as equally reliable or weigh them by the model's own confidence, since none uses the label to compare views; under full supervision, FCT does exactly that.

C. Loss Reweighting and Distillation

Training losses for segmentation often change how much each pixel counts. Focal loss [50] and online hard example mining [51] emphasize examples with a high loss, and the structure loss used by PraNet-style networks emphasizes pixels near boundaries [52]. By our identity, the consistency term of FCT belongs to this family: it reweights the supervised loss of each pixel, by how the pixel's error in one view compares with its error in another view of the same image.

A consistency term toward a detached prediction also resembles knowledge distillation, which trains a student to match the soft predictions of a teacher [53]. In self-distillation, the teacher is the model itself: an earlier generation of the same architecture [17], its own deeper layers [54], its predictions from the previous epoch [55] or dropout-sampled versions of itself [56]. Closest to our setting, data-distortion guided self-distillation [57] makes a network agree with itself on two distorted copies of an image under full supervision, treating both copies alike. Analyses of the distillation gradient show that it rescales the supervised gradient according to the teacher's confidence in the true class [17], [18], self-distillation over several rounds averages the labels of instances with correlated features [58], and Yuan et al. [19] relate distillation to label smoothing. Decoupled knowledge distillation [59] splits the loss exactly into target-class and non-target-class terms; knowledge adjustment [60], refined logits [61] and label revision [62] correct teacher predictions that contradict the label; and confidence-aware multi-teacher distillation [63] weights each teacher by how close its predictions are to the label. For segmentation, structured distillation transfers pixel-wise and pairwise knowledge from a large network to a compact one [64]. FCT is not a teacher–student scheme: it is one loss on the views of one image, within one network, and it neither corrects a target nor ranks separate teachers, but weights each pixel's consistency term by comparing the errors of two views with the label.

III. Consistency as Reweighted Supervision

This section analyses the consistency term of a multi-view training loss. It shows that, when the label is binary, a consistency term that draws one view toward the detached prediction of another acts at every pixel as the supervised loss, rescaled by a factor that compares the errors of the two views. The sign of this factor tells, pixel by pixel, whether the term helps or hurts, and Section IV builds the weights of FCT on it.

A. Notation

Let $x \in \mathbb{R}^{H\times W\times 3}$ denote an image, $y \in \{0,1\}^{H\times W}$ its polyp mask, and f a segmentation network that maps x to a logit map $z = f(x) \in \mathbb{R}^{H\times W}$; the predicted probabilities are $p = \sigma(z)$, where σ denotes the logistic function. Let T_h and T_v denote the horizontal and vertical flips. Each flip is its own inverse, so the prediction for a flipped view, brought back to the upright frame, is

$$z^k = T_k f(T_k x), \qquad k \in \{h, v\}, \tag{1}$$

and the upright view gives $z^o = f(x)$. A flip-equivariant network satisfies $z^h = z^v = z^o$. The flip gap of Section I measures departures from this equality on binary masks:

$$\Delta(x) = 1 - \frac{1}{2} \sum_{k\in\{h,v\}} D\big(\mathbf{1}[p^o > \tfrac{1}{2}],\, \mathbf{1}[p^k > \tfrac{1}{2}]\big), \tag{2}$$

where D denotes the Dice overlap of two masks and $\mathbf{1}[\cdot]$ the indicator function, applied pixel by pixel.

At a pixel i, the error of a view is $e_i = |p_i - y_i| \in [0, 1]$, and the view is correct at the decision threshold of $\frac{1}{2}$ if $e_i < \frac{1}{2}$. We drop the index i when a statement holds for each pixel separately. A consistency term draws a view s toward the prediction p_t of a target view t, which is detached, that is, treated as a constant. We write

$$\ell(z_s; q) = -q \log \sigma(z_s) - (1 - q) \log \big(1 - \sigma(z_s)\big) \tag{3}$$

for the binary cross-entropy of the view's logit z_s with a target $q \in [0, 1]$. The consistency term is $\ell(z_s; p_t)$ and the supervised loss is $\ell(z_s; y)$, and their gradients with respect to the logit are

$$\frac{\partial \ell(z_s; p_t)}{\partial z_s} = p_s - p_t, \qquad \frac{\partial \ell(z_s; y)}{\partial z_s} = p_s - y. \tag{4}$$

B. Consistency as an Adaptive Pixel Weight

The two gradients in Eq. (4) are parallel whenever the label is binary: the consistency gradient is the supervised one, rescaled by how the errors of the two views compare.

Proposition 1: Let $y \in \{0, 1\}$, $p_s, p_t \in [0, 1]$ and $p_s \neq y$. Then

$$\frac{\partial \ell(z_s; p_t)}{\partial z_s} = \Big(1 - \frac{e_t}{e_s}\Big) \frac{\partial \ell(z_s; y)}{\partial z_s}, \tag{5}$$

where $e_s = |p_s - y|$ and $e_t = |p_t - y|$ denote the errors of the view and of its target.

Proof: Because y is 0 or 1, every $p \in [0, 1]$ satisfies $p - y = (1 - 2y)|p - y|$: for $y = 0$, $p - y = p$, and for $y = 1$, $p - y = -(1 - p)$. Hence $p_s - p_t = (p_s - y) - (p_t - y) = (1 - 2y)(e_s - e_t) = (p_s - y)(1 - e_t/e_s)$, where the last step uses $(1 - 2y)e_s = p_s - y$ and $e_s > 0$. Eq. (4) completes the proof. ∎

The identity needs no assumption on the network or on the target's probability, only a binary label, which places both errors on the same side of the target. It also holds for a squared-error term on probabilities, whose gradient with respect to p_s is again $p_s - p_t$. Summed over pixels and differentiated with respect to the network parameters θ, Eq. (5) reads

$$\nabla_\theta \sum_i \ell(z_{s,i}; p_{t,i}) = \sum_i \omega_i \, \nabla_\theta \, \ell(z_{s,i}; y_i), \qquad \omega_i = 1 - \frac{e_{t,i}}{e_{s,i}}, \tag{6}$$

with the weights ω_i held constant. A consistency term is thus a supervised loss with per-pixel weights, and some of these weights are negative.

When the view is also supervised, the two terms add. With a cross-entropy supervised term of weight β and a consistency term of weight γ, the gradient on the view's logit is

$$\big(\beta + \gamma \, \omega\big) \, (p_s - y), \tag{7}$$

so the consistency term changes only how much each pixel's supervised gradient counts. Under full supervision, it brings no information beyond the label: it acts as an adaptive pixel weight, as focal loss [50] and the structure loss [52] do, but set by how the pixel's error in one view compares with its error in another.

Corollary 1 (Lag condition): Under the conditions of Proposition 1, a gradient step on $\ell(z_s; p_t)$ moves the view's probability toward the label if $e_t < e_s$, away from it if $e_t > e_s$, and not at all if $e_t = e_s$.

Proof: A step of size η changes the logit by $-\eta(p_s - p_t) = -\eta \, \omega \, (p_s - y)$, by Eq. (5). The supervised step $-\eta(p_s - y)$ moves p_s toward y because σ is increasing, and $\omega > 0$ if and only if $e_t < e_s$. ∎

To first order, the step lowers the view's error by $\eta \, \sigma'(z_s)(e_s - e_t)$, so we call $e_s - e_t$ the movement toward the label. Fig. 2(a) maps it over the two errors. Below the diagonal, the view lags behind its target, and the term helps. Above it, the view leads, and the term hurts, in two ways. Where the target is wrong and the view right ($e_s < \frac{1}{2} < e_t$), the term drives a correct pixel toward a misclassification, with a strength $e_t - e_s$ that can approach that of a full supervised step. Where both are right ($e_s < e_t < \frac{1}{2}$), the term only lowers the view's confidence, and it stops when the view reaches its target, on the correct side of the threshold.

In semi-supervised learning, no label separates the two cases, so the views are drawn together alike. Under full supervision, the label separates them at every pixel: the term can keep its full weight where the view lags behind its target, and elsewhere act only where it cannot reverse a correct decision. Section IV turns this rule into the weights of the FCT loss.

IV. Flip-Consistent Training

FCT is a training loss. It replaces the supervised loss of a segmentation network with three parts: the supervised loss on three orientation views of each image, a consistency term that draws the predictions of the views together, and label-aware weights that decide, pixel by pixel, how much each view may move toward another (Fig. 3). The network and its inference are unchanged. This section describes the three parts and the full loss, and then relates FCT to other losses.

A. Multi-View Supervision

Each image is passed through the network upright and flipped horizontally and vertically, and the flipped outputs are flipped back to the upright frame as in Eq. (1). Every view is supervised with the host network's own loss:

$$\mathcal{L}_{\text{sup}} = \frac{1}{3} \sum_{k \in \{o,h,v\}} \mathcal{L}_{\text{host}}\big(Z^k, y\big), \tag{8}$$

where Z^k denotes all output maps of view k in the upright frame and $\mathcal{L}_{\text{host}}$ the supervised loss of the host network, unchanged; the prediction logit z^k is formed from Z^k as the host forms it at test time. The factor $\frac{1}{3}$ keeps the supervised loss at the scale of single-view training. On its own, Eq. (8) is a deterministic form of flip augmentation: each view is tied to the label, but no view is tied to another.

B. Cross-View Consistency

The consistency term ties the views to each other. It draws each flipped view k toward the detached upright prediction,

$$\mathcal{C}_k = \frac{1}{HW} \sum_i w_i^k \, \ell\big(z_i^k; p_i^o\big), \tag{9}$$

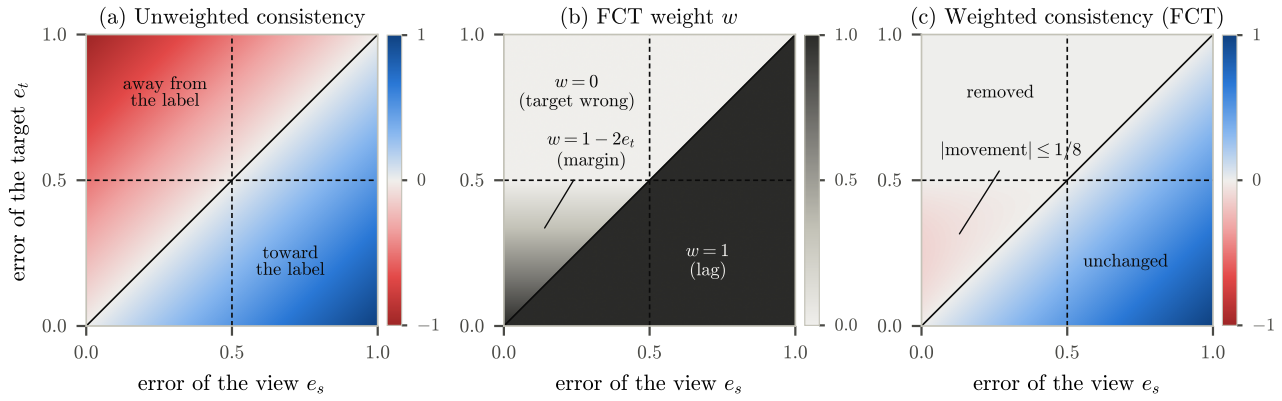


Fig. 2. Effect of a consistency term on the error of a view at one pixel, as a function of the view's error e_s and the error e_t of its detached target (Proposition 1); dashed lines mark the decision threshold. (a) Without a weight, the term moves the view away from the label wherever the view is the closer of the two, even toward a wrong target (top left). (b) The FCT weight of Eq. (11). (c) With it, every movement toward the label is kept, every movement away from the label toward a wrong target is removed, and the remaining movements away from the label are bounded by $1/8$.

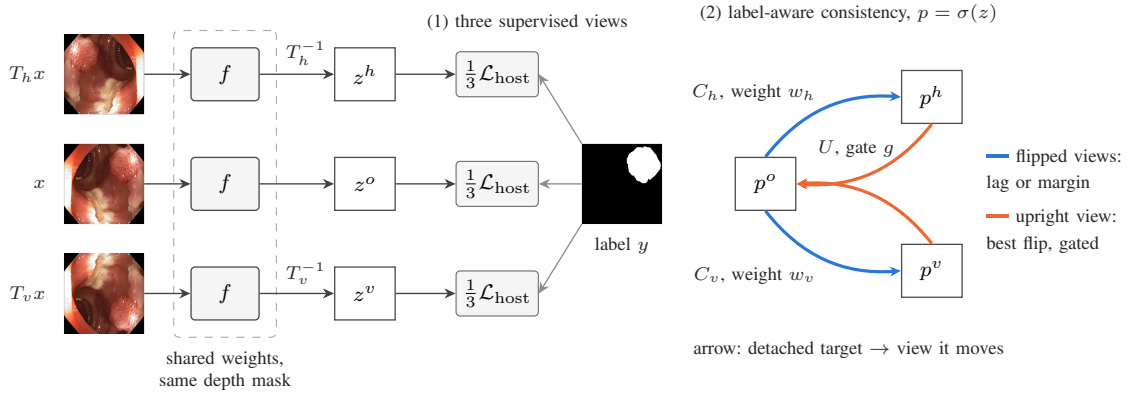


Fig. 3. Overview of the FCT loss. (1) An image and its horizontal and vertical flips pass through one network; the flipped outputs are flipped back, and each view is supervised with the host network's own loss, weighted $\frac{1}{3}$. (2) A consistency term draws the predictions of the views together, each toward a detached target: the flipped views toward the upright view, with the lag-or-margin weight of Eq. (11), and the upright view toward the flipped view closest to the label, wherever that view is more accurate. At test time, only the upright view is used.

with the weights w_i^k of Section IV-C, and it draws the upright view toward the flipped view closest to the label, wherever that view is more accurate:

$$\mathcal{U} = \frac{1}{HW} \sum_i g_i \, \ell\big(z_i^o; p_i^*\big), \qquad g_i = \mathbf{1}\big[e_i^* < e_i^o\big], \tag{10}$$

where p_i^* denotes the detached probability of the flipped view with the smaller error at pixel i, and e_i^* its error. Together, the two directions let the more accurate side move the other: where the upright view is more accurate, the flipped views move toward it, and where a flipped view is more accurate, the upright view, which is the one used at test time, moves toward it. By Corollary 1, every movement that the gate g allows in Eq. (10) is toward the label.

C. Label-Aware Weights

The weights decide how much each flipped view moves toward the upright view. Following Corollary 1, the term keeps its full weight where the flipped view lags behind the upright view, and it is weakened where the flipped view leads:

$$w_i^k = \begin{cases} 1, & e_i^k \ge e_i^o, \\ \max\big(0,\, 1 - a\, e_i^o\big), & e_i^k < e_i^o, \end{cases} \tag{11}$$

where e_i^k and e_i^o denote the errors of the flipped and the upright view at pixel i, computed without gradient, and $a > 0$ is a constant.

The constant a follows from the decision threshold. On a leading pixel, the term moves the flipped view away from the label (Corollary 1), and Eq. (11) keeps it in proportion to the margin $1 - a\, e^o$. We ask two things of this margin. First, a movement toward a wrong target must vanish, $w = 0$ whenever $e^o > \frac{1}{2}$, which requires $a \ge 2$. Second, a movement toward a correct target must survive however small its margin, $w > 0$ whenever $e^o < \frac{1}{2}$, which requires $a \le 2$. Hence $a = 2$, and on a leading pixel with a correct target, $w = 1 - 2e^o = |2p^o - 1|$ is twice the target's distance from the threshold, that is, its margin rescaled to $[0, 1]$. The constant is thus set by the

threshold rather than tuned, and Section V-G shows that the results are insensitive to it.

With $a = 2$, the term on a leading pixel is weak. Its strength $w\,(e^o - e^k)$ is at most $(1 - 2e^o)\,e^o \le \frac{1}{8}$, against up to one for a supervised step (Fig. 2(c)), and it vanishes when the flipped view reaches the upright prediction, on the correct side of the threshold. It thus keeps the two views consistent where both are right, as flip equivariance asks, without reversing a decision by itself.

D. Full Loss and Training

The FCT loss adds the consistency term to the multi-view supervision:

$$\mathcal{L}_{\mathrm{FCT}} = \mathcal{L}_{\mathrm{sup}} + \frac{\lambda}{2}\big(\mathcal{C}_h + \mathcal{C}_v\big) + \frac{\lambda}{2}\,\mathcal{U}, \tag{12}$$

where $\lambda \ge 0$ weighs consistency against supervision. The two flipped views share λ, and the upright view's term receives the weight of one flipped view's. The weight is ramped up linearly at the start of training,

$$\lambda_t = \lambda \,\min\big(1,\, t/t_0\big), \tag{13}$$

where t counts the training batches and t_0 is the length of the ramp; we use $\lambda = 0.5$ and $t_0 = 300$.

Three implementation choices keep the views comparable and the memory low. First, the three views of an image share one stochastic-depth mask [65], so that their differences reflect the flip rather than the random depth. Second, when a host trains at several input scales, FCT is applied at every scale. Third, the probabilities p^h and p^v of Eq. (10) come from passes without gradient that leave the running statistics of batch normalization unchanged. In our current implementation, these passes run before the upright pass and the graph of each view is released after its backward pass, so only one view's computation graph is held in memory at a time; the runs reported in Section V computed them after the upright pass, which gives the same loss at a higher peak memory. For each image and input scale, FCT thus runs five forward passes, three of them with gradients. At test time, the network is applied to the upright image only, exactly as without FCT, so FCT changes neither the architecture nor its inference cost.

E. Relation to Other Losses

By Eq. (7), FCT is a supervised loss with adaptive pixel weights: on a flipped view, the supervised weight $\frac{1}{3}$ is raised where the view lags behind the upright view and slightly lowered where it leads. This places FCT among losses that reweight pixels. Focal loss [50] and online hard example mining [51] weight a pixel by its own error, and the structure loss of PraNet-style networks [52] by its distance to a boundary; FCT weights it by how its error compares with that of another view of the same pixel.

The weights of Eq. (11) also contain the usual consistency rules as special cases or near neighbours, which Section V-G compares. With $w = 1$ everywhere, $\mathcal{C}_k$ is the unweighted transformation consistency of semi-supervised segmentation [13], [14], applied here to labeled images; by Eq. (6), it mixes positive and negative pixel weights. Confidence thresholding [12] keeps the terms whose target is confident, which includes confidently wrong targets, the case that Corollary 1 rules out. Finally, a rule can use the label to judge the target alone, for example by weighting each pixel by whether and how confidently the upright view is correct. Such a rule ignores the error of the view being moved, so it cannot tell the lagging pixels, where the term helps, from the leading ones, where it hurts. Of these rules, only FCT compares the two errors, as Corollary 1 requires.

V. Experiments

We test the claims of Sections III and IV in the order in which they were made: FCT improves all three networks over their own baselines, most on unseen data (Section V-C), and a label-aware variant generalizes to six more centres (Section V-D); FCT makes predictions more flip-consistent (Section V-E); a per-image analysis shows where its gains come from (Section V-F); and a study of the loss terms shows that the multi-view supervision gives most of the gain, while the label-aware weights shape the gradient as Corollary 1 predicts and need no tuning (Section V-G). Section V-H reports its cost.

A. Datasets and Evaluation Protocol

We train on 1,288 images, 800 from Kvasir-SEG [26] and 488 from CVC-ClinicDB [27], drawn from the 1,450-image training split of [5]; none of them appears in a test set. Because this set is smaller than the full split, absolute scores are not directly comparable with published ones, and every comparison below is between models trained on the same images. We test on the five standard test sets: the test splits of Kvasir-SEG (100 images) and CVC-ClinicDB (62), which we call seen, and CVC-ColonDB [28] (380), CVC-300 [29] (60) and ETIS [30] (196), which are unseen in training. CVC-ColonDB and CVC-300 come from the same group as CVC-ClinicDB, whereas ETIS comes from another centre.

We report the mean Dice (mDice) of the evaluation toolbox of [5], which Polyp-PVT also uses: each predicted map is min-max normalized and binarized at 256 thresholds, and the Dice coefficient is averaged over images and thresholds. The flip gap of Eq. (2) is computed on the masks thresholded at 0.5 at the input resolution of the network, with add-one smoothing so that two empty masks agree. Each network is evaluated on the five test sets after every epoch with the per-image Dice of the official Polyp-PVT code, and we keep the epoch with the highest five-set mean, for vanilla and FCT alike; Section V-C also reports the gains without this selection.

Each configuration is trained with six random seeds, and vanilla and FCT share each seed. We report the mean and standard deviation (SD) over seeds, the mean paired difference with its 95% confidence interval (CI) from the t-distribution with five degrees of freedom, and the number of seeds on which FCT scores higher. An exact permutation test over all 924 ways of splitting the twelve runs of a comparison into two groups of six reaches the same conclusion as the CI for every entry of Tables I and II.

B. Implementation Details

We add FCT to three networks without changing their architectures or retuning their recipes. Polyp-PVT [7] uses a PVTv2-B2 encoder [66]. As in its official code, it is trained with AdamW, a learning rate of 10^{-4}, a weight decay of 10^{-4}, batches of 16 images of 352×352 pixels, input scales of 0.75, 1 and 1.25, element-wise gradient clipping at 0.5 and 99 epochs, and its encoder has a stochastic-depth rate of 0.1; we add mixed precision. Its official recipe applies no orientation augmentation by default, and we keep this default, so vanilla Polyp-PVT never sees a flipped image. PraNet-V2 [8] uses a Res2Net-50 encoder [67], which has no stochastic depth, and is trained with Adam, a constant learning rate of 10^{-4}, batches of 8 images and 29 epochs, with the data pipeline, input scales and mixed precision of Polyp-PVT. ASGNet [9] uses a PVTv2-B3 encoder and keeps its own recipe: Adam, batches of 40 images accumulated over eight parts, 79 epochs at a single scale in full precision, a cosine learning-rate schedule, and its own data loader, which applies random horizontal flips, crops, rotations, colour changes and label noise. Unlike the other two, ASGNet is therefore trained with flip augmentation even without FCT. FCT uses $\lambda = 0.5$, set in preliminary runs on Polyp-PVT, and $a = 2$ for all three networks. Each run uses one [GPU model] GPU.

C. Main Results

FCT improves the five-set mean Dice of Polyp-PVT and PraNet-V2 on every seed and that of ASGNet on five of six, and for every network its largest gain is on an unseen set (Tables I–III, Fig. 4). For Polyp-PVT, the five-set mean rises by 0.52 points, with a 95% CI of $[0.15, 0.90]$, and ETIS gains 1.65 points $[0.88, 2.42]$ on all six seeds; the gains on the other four sets are positive on average but within seed variation. For PraNet-V2, the five-set mean rises by 1.68 points $[0.69, 2.67]$. Its largest gains are on ETIS, 3.61 points $[0.23, 6.99]$, and on CVC-ColonDB, 3.01 points $[1.25, 4.77]$ on all six seeds, and Kvasir also improves, by 1.69 points $[0.12, 3.26]$. For PraNet-V2, FCT also lowers the SD over seeds on every set. For these two networks, no set loses on average; the smallest gains are on CVC-300 for both, $+0.06$ and $+0.04$ points, and on CVC-ClinicDB for PraNet-V2, $+0.03$. ASGNet, whose own recipe already trains with random horizontal flips, gains least (Table III). Its ETIS score rises by 1.69 points $[0.08, 3.30]$ on five of six seeds, but its five-set mean rises by only 0.31 points: the CI excludes zero by less than 0.01 points (paired $p = 0.049$), and the exact permutation test does not support the gain ($p = 0.20$). Its other four sets change by less than 0.6 points in either direction, within seed variation.

Without epoch selection, the gains remain positive on average but are less certain. Fig. 5 shows the five-set mean of the training-time evaluator after every epoch: the six-seed mean of FCT lies above that of vanilla after 81 of 99 epochs for Polyp-PVT, 28 of 29 for PraNet-V2 and 76 of 79 for ASGNet. Averaged over the last ten epochs, without any selection, FCT gains $+0.42$ $[-0.03, 0.88]$, $+2.41$ $[1.40, 3.43]$ and $+0.21$ $[-0.59, 1.01]$ points on the five-set mean of the three networks, and $+1.52$ $[0.11, 2.93]$, $+3.71$ $[0.75, 6.68]$ and $+1.99$ $[-0.26, 4.24]$ points on ETIS (Appendix, Table XI). At the last epoch alone, no five-set mean gain has a CI above zero: $+0.55$ $[-0.11, 1.22]$ for Polyp-PVT, $+0.49$ $[-1.17, 2.14]$ for PraNet-V2, whose curves peak early and decline, so that its last epoch is a poor checkpoint for either arm, and $+0.13$ $[-0.91, 1.17]$ for ASGNet.

TABLE I
POLYP-PVT (PVTV2-B2) WITH AND WITHOUT FCT ON FIVE POLYP TEST SETS: MDICE (%), MEAN ± SD OVER SIX SEEDS. BOLD: GAINS WHOSE 95% CI EXCLUDES ZERO.

Test set	Vanilla	FCT (ours)	Gain	95% CI	Seeds
Seen					
Kvasir	91.94 ± 0.59	92.50 ± 0.41	+0.56	[−0.31, 1.44]	4/6
CVC-ClinicDB	93.17 ± 0.64	93.36 ± 0.23	+0.19	[−0.64, 1.01]	4/6
Unseen					
CVC-ColonDB	80.28 ± 0.68	80.43 ± 0.60	+0.15	[−0.80, 1.10]	3/6
CVC-300	88.79 ± 0.86	88.85 ± 1.10	+0.06	[−1.39, 1.51]	2/6
ETIS	78.61 ± 0.29	80.26 ± 0.65	**+1.65**	**[0.88, 2.42]**	6/6
Mean	86.56 ± 0.17	87.08 ± 0.32	**+0.52**	**[0.15, 0.90]**	6/6

Training set: 1,288 Kvasir and CVC-ClinicDB images; *seen*: test splits of the two training datasets; *unseen*: datasets not used in training. Gain: mean paired difference, FCT minus vanilla with the same seed. 95% CI: t-distribution, five degrees of freedom. Seeds: seeds, of six, on which FCT scores higher. Mean: average over the five test sets.

TABLE II
PRANET-V2 (RES2NET-50) WITH AND WITHOUT FCT ON FIVE POLYP TEST SETS: MDICE (%), MEAN ± SD OVER SIX SEEDS. BOLD: GAINS WHOSE 95% CI EXCLUDES ZERO.

Test set	Vanilla	FCT (ours)	Gain	95% CI	Seeds
Seen					
Kvasir	87.94 ± 1.46	89.63 ± 0.55	**+1.69**	**[0.12, 3.26]**	5/6
CVC-ClinicDB	89.80 ± 1.30	89.83 ± 0.86	+0.03	[−1.31, 1.38]	3/6
Unseen					
CVC-ColonDB	69.91 ± 2.30	72.92 ± 1.27	**+3.01**	**[1.25, 4.77]**	6/6
CVC-300	87.73 ± 2.92	87.76 ± 1.47	+0.04	[−3.31, 3.39]	3/6
ETIS	67.07 ± 1.72	70.68 ± 1.69	**+3.61**	**[0.23, 6.99]**	5/6
Mean	80.49 ± 0.93	82.17 ± 0.64	**+1.68**	**[0.69, 2.67]**	6/6

Columns and statistics as in Table I.

D. Six More Centres

Of the five standard test sets, only ETIS comes from another centre than the training images (Section V-A). PolypGen [1] adds six more centres; we use its 1,411 single-frame images with a polyp, none of which took part in training or in selecting a checkpoint. We fixed the analysis before scoring any map: the primary result is the mDice pooled over the six centres, every image counted once, over all six seeds, and each centre is a secondary result. So far, PolypGen maps exist only for the confidence-graded weight, row (h) of Table VII, and not yet for FCT. This label-aware variant raises the pooled mDice of Polyp-PVT from 82.14% to 82.98%, by 0.84 points $[0.59, 1.09]$, on all six seeds, and its gain has a CI above zero on four of the six centres (Table IV); an exact permutation test over the 924 splits of the twelve runs gives $p = 0.002$ for the pooled gain. FCT's checkpoints will be scored with the same analysis.

TABLE III
ASGNET (PVTV2-B3) WITH AND WITHOUT FCT ON FIVE POLYP TEST SETS: MDICE (%), MEAN ± SD OVER SIX SEEDS. BOLD: GAINS WHOSE 95% CI EXCLUDES ZERO.

Test set	Vanilla	FCT (ours)	Gain	95% CI	Seeds
Seen					
Kvasir	90.86 ± 0.15	90.80 ± 0.33	−0.06	[−0.46, 0.34]	1/6
CVC-ClinicDB	91.09 ± 1.11	90.87 ± 0.75	−0.21	[−0.96, 0.53]	2/6
Unseen					
CVC-ColonDB	80.41 ± 0.71	80.98 ± 1.04	+0.57	[−0.72, 1.85]	4/6
CVC-300	89.40 ± 0.41	88.96 ± 0.97	−0.44	[−1.50, 0.62]	3/6
ETIS	77.44 ± 1.23	79.12 ± 0.85	**+1.69**	**[0.08, 3.30]**	5/6
Mean	85.84 ± 0.42	86.15 ± 0.33	+0.31	[0.00, 0.61]	5/6

Columns and statistics as in Table I. ASGNet's own recipe already trains with random horizontal flips. Its five-set mean gain is borderline: the CI excludes zero by less than 0.01 points (paired t-test, $p = 0.049$), and the exact permutation test gives $p = 0.20$.

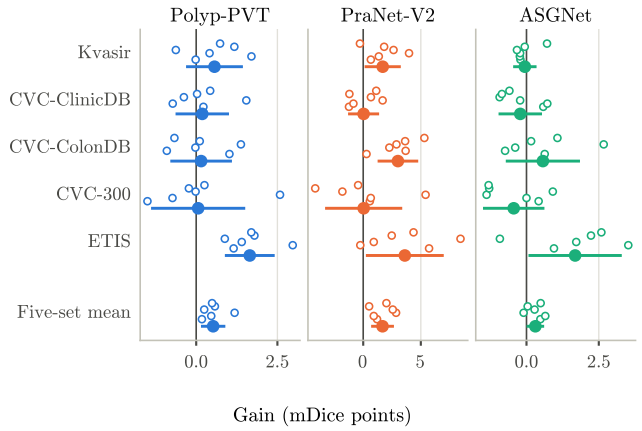


Fig. 4. Paired gains of FCT over vanilla in mDice points: one hollow marker per seed, and the mean with its 95% CI below them. Every seed gains on the five-set mean for Polyp-PVT and PraNet-V2 and five of six for ASGNet; the largest gains are on ETIS and, for PraNet-V2, CVC-ColonDB.

E. Flip Consistency

FCT makes both networks more flip-consistent on every test set (Table V). It lowers the five-set mean flip gap from 12.5% to 7.5% for Polyp-PVT and from 15.6% to 10.2% for PraNet-V2, and the flip gap on the training images by 84% and 76%. The reduction holds on every seed and in every row of the table. Two observations qualify it. First, flip consistency is not accuracy: the largest reduction on a test set, 70% for Polyp-PVT on CVC-ClinicDB, comes with almost no change in Dice, and CVC-ColonDB and ETIS remain the least consistent sets after FCT. Second, training on flipped images lowers the flip gap by itself. In preliminary single runs of Polyp-PVT, the multi-view supervision of row (b) of Table VII, without the consistency term, reached a five-set flip gap of 8.2%, and the official random flip and rotation augmentation reached 6.8%, below that of FCT; yet the five-set mean Dice of the latter, 85.75%, was below that of every vanilla seed. A lower flip gap alone therefore does not explain the gains of FCT, and we read it as a symptom that the views are tied together, not as the cause of the gains.

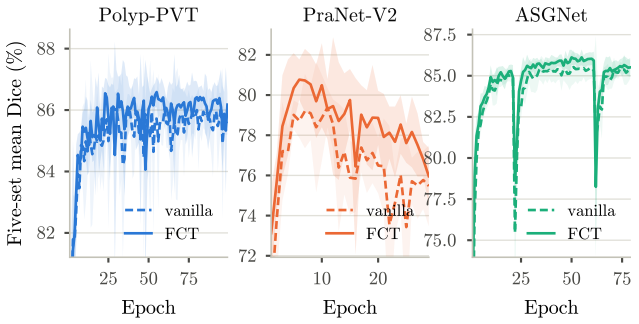


Fig. 5. Five-set mean Dice of the training-time evaluator after every epoch, without (dashed) and with (solid) FCT; mean and SD over six seeds. FCT lies above vanilla through most of training. Every ASGNet run, with or without FCT, drops briefly at epoch 62, so the drop comes from its recipe, not from FCT.

TABLE IV
SIX MORE CENTRES (POLYPGEN): MDICE (%) OF VANILLA POLYP-PVT AND OF ITS CONFIDENCE-GRADED VARIANT, ROW (H) OF TABLE VII, MEAN ± SD OVER SIX SEEDS. BOLD: GAINS WHOSE 95% CI EXCLUDES ZERO.

Centre	Images	Vanilla	Variant (h)	Gain [95% CI]	Seeds
C1	251	86.65 ± 0.42	87.94 ± 0.55	**+1.29 [0.71, 1.86]**	6/6
C2	270	84.60 ± 0.47	84.31 ± 0.64	−0.28 [−1.27, 0.70]	2/6
C3	455	89.93 ± 0.24	90.46 ± 0.20	**+0.53 [0.25, 0.81]**	6/6
C4	146	68.33 ± 0.99	69.23 ± 1.47	+0.90 [−1.63, 3.44]	4/6
C5	206	65.71 ± 1.65	67.92 ± 1.25	**+2.21 [0.38, 4.04]**	5/6
C6	83	82.88 ± 0.29	84.23 ± 0.45	**+1.35 [1.04, 1.65]**	6/6
Pooled	1411	82.14 ± 0.18	82.98 ± 0.21	**+0.84 [0.59, 1.09]**	6/6

Single-frame images with a polyp; none was used in training or in selecting a checkpoint. Pooled: every image of the six centres counted once, the primary result of an analysis fixed before scoring. FCT itself: pending.

F. Where Do the Gains Come From?

For PraNet-V2, the gains on the unseen sets are spread over many images: on ETIS, the 176 images outside the ten largest gains and the ten largest losses contribute 2.94 of its 3.61 points. For Polyp-PVT, they are more concentrated: on ETIS, the ten images with the largest gains contribute 1.78 points, more than the net gain of 1.65, and the other 176 images contribute 0.79. Table VI breaks the errors down at a threshold of 0.5 on the per-image min-max-normalized maps. The two networks gain in different ways. PraNet-V2 fails less often: the number of images with a Dice below 0.1 falls from 31.8 to 25.0 per run on ETIS and from 56.5 to 47.5 on CVC-ColonDB. Polyp-PVT fails about as often with FCT as without, 10.3 and 10.5 images per run on ETIS; it gains instead by predicting less false-positive area, 1.31 instead of 1.52 times the polyp area. In two cases, small polyps carry much of the gain: the quarter of images with the smallest polyps gains 5.46 points on ETIS for Polyp-PVT and 6.16 points on CVC-ColonDB for PraNet-V2. The pattern is not general: on ETIS, the smallest polyps gain 2.70 points for PraNet-V2 and the next quarter 6.05, and on CVC-300 the smallest polyps lose with both networks. These breakdowns are descriptive, and with four size groups per set and network, some differences are expected by chance.

Fig. 6 shows examples. Without FCT, PraNet-V2 misses

TABLE V
FLIP GAP (%) OF THE PREDICTED MASKS WITHOUT AND WITH FCT ON EACH TEST SET AND ON THE TRAINING IMAGES, AND ITS RELATIVE CHANGE. MEANS OVER SIX SEEDS; LOWER IS BETTER.

	Polyp-PVT			PraNet-V2		
Images	Vanilla	FCT	Change	Vanilla	FCT	Change
Seen						
Kvasir	6.3	4.1	−35%	8.8	5.1	−42%
CVC-ClinicDB	12.4	3.7	−70%	13.1	5.9	−55%
Unseen						
CVC-ColonDB	16.9	11.3	−33%	20.4	14.8	−27%
CVC-300	9.1	5.1	−43%	9.8	6.2	−37%
ETIS	17.8	13.2	−26%	25.8	19.2	−26%
Five-set mean	12.5	7.5	−40%	15.6	10.2	−34%
Training images	6.3	1.0	−84%	8.5	2.0	−76%

Flip gap: one minus the Dice overlap of the binary masks predicted for an image and for its flipped copy, flipped back, averaged over horizontal and vertical flips and over images. FCT lowers it on every seed, in every row.

TABLE VI
WHERE THE GAINS COME FROM ON THE UNSEEN TEST SETS, VANILLA → FCT: FAILURES, MISSED AND FALSE-POSITIVE AREA, AND THE GAIN ON THE SMALLEST POLYPS. MEANS OVER SIX SEEDS.

Test set	Gain	Failures	Missed area	False-pos. area	Smallest
Polyp-PVT					
CVC-ColonDB	+0.15	15.0 → 15.3	0.18 → 0.16	0.66 → 0.73	−1.75
CVC-300	+0.06	0.2 → 0.0	0.06 → 0.06	0.23 → 0.27	−2.38
ETIS	+1.65	10.5 → 10.3	0.12 → 0.11	1.52 → 1.31	+5.46
PraNet-V2					
CVC-ColonDB	+3.01	56.5 → 47.5	0.27 → 0.25	0.33 → 0.26	+6.16
CVC-300	+0.04	0.2 → 0.0	0.05 → 0.06	0.25 → 0.23	−1.85
ETIS	+3.61	31.8 → 25.0	0.26 → 0.22	0.82 → 0.86	+2.70

Gain: mDice points. Failures: images per run with Dice below 0.1 for the mask thresholded at 0.5 after per-image min-max normalization. Missed and false-positive area: at the same threshold, as fractions of the polyp area. Smallest: mDice gain on the quarter of the set with the smallest polyps.

the polyp in rows 1, 2 and 4, and Polyp-PVT adds spurious regions in rows 1–3; with FCT, both predictions concentrate on the polyp. The last row shows the image on which FCT loses most: with FCT, Polyp-PVT adds a false region, and PraNet-V2 misses the polyp.

G. Ablation Studies

Loss terms. The FCT loss has three parts: the multi-view supervision of Eq. (8), the consistency term of Eqs. (9) and (10), and the label-aware weights of Eq. (11). Table VII dissects them on Polyp-PVT, against the same vanilla runs throughout. Row (b) keeps only the multi-view supervision. It is also the flip-augmentation control, since it trains on exactly the views that FCT uses, with the same supervision. Rows (c)–(e) add the consistency term on the upright view, on the flipped views, and on both, which is FCT, so their differences from row (b) are the effect of the consistency term. Rows (f) and (g) change only the weights of FCT, to unweighted consistency and to the lag gate without its margin. We fixed the protocol before seeing these rows: rows (b) and (c) are trained with six seeds, and rows (f) and (g) with three seeds first, extended to six only if they land near FCT. Row (c) is being repeated after a failed run; rows (f) and (g) landed near FCT and are being extended to six seeds. With six seeds, the paired CI of the difference between two variants spans about ± 0.4 to ± 0.7 points on the five-set mean, and with three seeds about 2.4 times more, so small differences between variants are not resolved.

The multi-view supervision gives most of the gain on ETIS. On its own, row (b) raises ETIS by 1.48 points $[0.32, 2.65]$ on five of six seeds, against 1.65 points for FCT, and the five-set mean by 0.37 points $[-0.01, 0.74]$. This gain is not that of random flip augmentation: ASGNet already trains with random flips, and FCT still raises its ETIS score by 1.69 points (Table III). The consistency term and its weights complete the loss: FCT is the only variant that improves all five sets on average. The multi-view supervision alone lowers CVC-ClinicDB by 0.48 points, and FCT recovers it, although this difference is borderline, $+0.66$ $[-0.01, 1.34]$. The consistency term's own effect on the five-set mean, row (e) minus row (b), is $+0.16$ $[-0.30, 0.62]$ points, and the weight functions are within seed variation of one another: FCT minus rows (f) and (g) is $+0.29$ $[-0.87, 1.45]$ and $+0.29$ $[-1.65, 2.23]$ points on three seeds, and row (d) minus row (h) is $+0.06$ $[-0.60, 0.73]$. Row (h) uses the confidence-graded weight sketched in Section IV-E, which keeps a weight of 1 where the upright view is correct, raises it to $1 + p^o$ on missed polyp pixels and lowers it to $1 - p^o$ on false-positive pixels.

The weights shape the gradient of the consistency term, as Eq. (6) predicts. With the lag-based weights of rows (d), (e) and (g), the gradient of the term on the flipped views has a cosine of 0.53 to 0.56 with the gradient of the rest of the loss, and between 0.47 and 0.56 on every seed. Without weights, row (f), it falls to 0.23, and with the confidence-graded weight, row (h), to 0.26; these two stay between 0.18 and 0.33 on every seed (Fig. 7). Unweighted and confidence-graded terms keep full weight on leading flipped views, whose pixel weights in Eq. (6) are negative, so part of their gradient points away from the label. Leading pixels are common: in the runs where we logged them, Polyp-PVT with $a < 2$ and PraNet-V2, a flipped view leads the upright view on about 44% of its pixels at the end of training, although this is a share of pixels, not of the gradient. The ratio of the two gradients' norms follows the same order: 0.07 to 0.08 with lag-based weights, against 0.11 without weights and 0.12 with the graded weight.

Constant a. The results are insensitive to a, so the constant needs no tuning (Table VIII; row (g) of Table VII adds the limit $a \to \infty$, which keeps only the lagging terms). Every value in $\{0.5, 1, 1.5, 2\}$ improves ETIS by 1.6 to 2.1 points, and none differs significantly from $a = 2$ on the five-set mean. The value $a = 2$ is the only one that improves all five sets on average, but this distinction is within seed variation. These runs do not fully explain the insensitivity. The case that $a = 2$ removes, a flipped view that leads a wrong upright view, is rare: it arises on only 0.16% of the pixels at the end of training (Polyp-PVT, $a < 2$) and on 0.24% (PraNet-V2). But a also scales the term on the far more frequent leading pixels with a correct upright view, and this change, too, leaves Dice unchanged within seed variation.

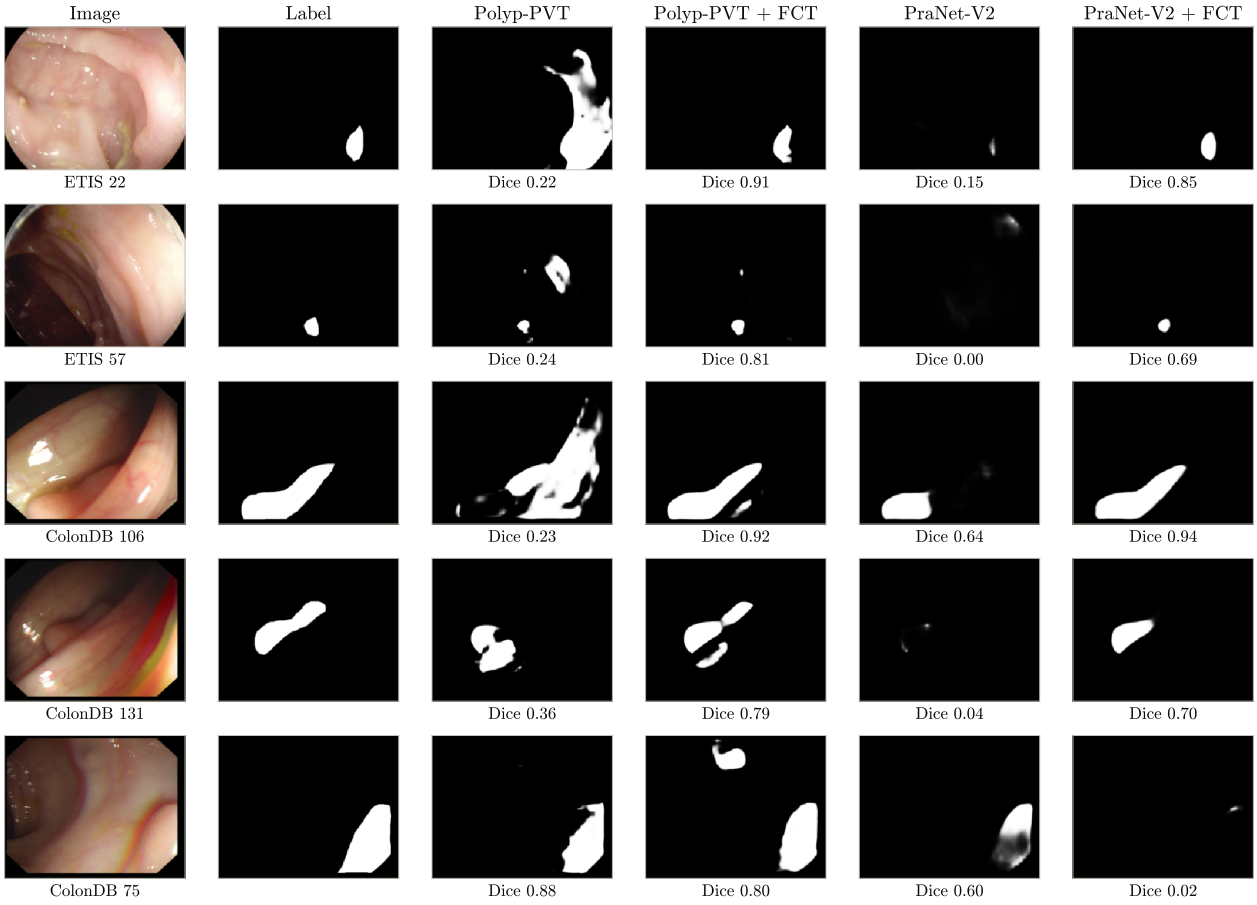


Fig. 6. Examples from the unseen sets: image, label, and the maps predicted by each network without and with FCT (seed 1), each with its mDice. Rows 1–4: four of the twelve unseen-set images with the largest gains, averaged over six seeds and both networks. Row 5: the image with the largest loss.

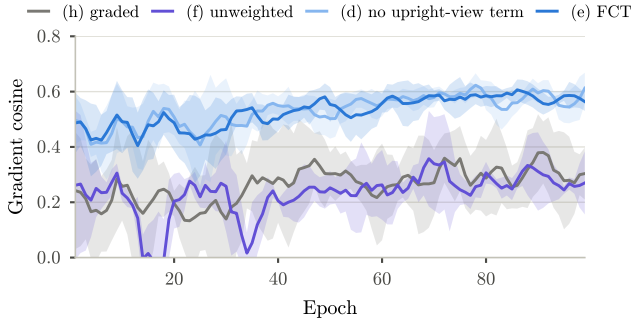


Fig. 7. Cosine between the gradient of the consistency term on the flipped views and that of the rest of the loss during training of Polyp-PVT, for rows (d), (e), (f) and (h) of Table VII; per-epoch means, smoothed over five epochs, with the SD over seeds. The lag-based weights double the alignment of the consistency gradient with the rest of the loss.

H. Cost

FCT multiplies the training time per epoch by 3.6 for Polyp-PVT, 3.3 for PraNet-V2 and 3.7 for ASGNet (Table IX). This matches its five forward and three backward passes for each image and input scale, which cost $11/3 \approx 3.7$ times a single view if a backward pass costs twice a forward pass. Without the term on the upright view, the two passes without gradient are not needed, and the measured factor drops to 3.0. Inference is unchanged: the trained network is the host network, applied once to the upright image.

VI. Discussion and Limitations

What each part of the loss does. The study of the loss terms separates two roles. The multi-view supervision carries most of the accuracy gain on the unseen ETIS set. The consistency term and its label-aware weights complete the loss: with them, FCT is the only variant that improves all five test sets on average, and the consistency gradient is twice as well aligned with supervision as without the weights. By Eq. (7), this division of roles is expected: under full supervision, the consistency term brings no information beyond the label, only a pixel weight, so its effect on accuracy is that of a reweighting of the supervised loss, and its weights decide whether that reweighting ever turns against the label.

Why the unseen sets? For all three networks, FCT gains most on ETIS, the test set from another centre, and, for PraNet-V2, it also gains on CVC-ColonDB. These are the two sets on which every network is least accurate, and on which Polyp-PVT and PraNet-V2 are least flip-consistent. One explanation is the hypothesis of Section I: training on the flipped views

TABLE VII
ABLATION OF THE FCT LOSS TERMS ON POLYP-PVT: PAIRED MDICE GAINS (POINTS, 95% CI), THE EFFECT OF EACH TERM, TEST SETS IMPROVED, FLIP GAP AND GRADIENT ALIGNMENT. BOLD: CIS THAT EXCLUDE ZERO.

		Consistency term							
Variant	Views	Flipped	Upright	Mean gain	ETIS gain	Effect	Sets	Flip gap (%)	Cosine
Building the FCT loss on the multi-view supervision									
(a) Vanilla	–	–	–	–	–	–	–	12.5	–
(b) Multi-view supervision	✓	–	–	+0.37 [−0.01, 0.74]	**+1.48 [0.32, 2.65]**	–	3/5	–	–
(c) + upright-view term	✓	–	✓	pending	pending	pending	–	–	–
(d) + flipped-view term	✓	Eq. (11)	–	+0.47 [−0.02, 0.96]	**+1.97 [1.06, 2.87]**	+0.10 [−0.45, 0.66]	4/5	7.5	0.53
(e) + both (FCT)	✓	Eq. (11)	✓	**+0.52 [0.15, 0.90]**	**+1.65 [0.88, 2.42]**	+0.16 [−0.30, 0.62]	5/5	7.5	0.53
Changing the weights of FCT									
(f) FCT, unweighted (3 seeds)	✓	$w = 1$	✓	+0.32 [−0.50, 1.14]	+1.47 [−1.40, 4.34]	+0.29 [−0.87, 1.45]	3/5	–	0.23
(g) FCT, lag only (3 seeds)	✓	lag only	✓	+0.32 [−0.45, 1.09]	+0.94 [−0.07, 1.94]	+0.29 [−1.65, 2.23]	4/5	–	0.56
Another label-aware weight, without the upright-view term									
(h) Graded weight	✓	graded	–	**+0.40 [0.13, 0.68]**	**+1.56 [0.53, 2.60]**	+0.06 [−0.60, 0.73]	3/5	7.7	0.26

Six seeds unless the row says otherwise. Mean and ETIS gain: over vanilla with the same seeds. Effect, on the five-set mean, paired by seed: for rows (c)–(e), the row minus row (b), the effect of the consistency term added to the multi-view supervision; for rows (f) and (g), FCT minus the row, the effect of FCT's weights; for row (h), row (d) minus row (h). Sets: test sets, of five, with a positive mean gain. Cosine: between the gradient of the consistency term on the flipped views and that of the rest of the loss, measured every 50 batches and averaged over training and seeds; for rows (e)–(g), the rest includes the upright-view term. Lag only: $w = 1$ where the flipped view lags and 0 where it leads ($a \to \infty$). Graded: the confidence-graded weight of Section V-G.

TABLE VIII
SENSITIVITY TO THE CONSTANT a OF EQ. (11) ON POLYP-PVT: PAIRED MDICE GAINS OVER VANILLA AND OVER $a = 2$ (POINTS, SIX SEEDS, 95% CI). BOLD: CIS THAT EXCLUDE ZERO.

a	Mean gain	ETIS gain	Sets	$a = 2$ minus row
0.5	**+0.51 [0.30, 0.73]**	**+2.15 [1.37, 2.92]**	3/5	+0.01 [−0.47, 0.49]
1	+0.33 [−0.02, 0.67]	**+1.58 [0.40, 2.77]**	3/5	+0.19 [−0.26, 0.65]
1.5	**+0.36 [0.02, 0.71]**	**+1.83 [0.83, 2.82]**	3/5	+0.16 [−0.24, 0.56]
2 (FCT)	**+0.52 [0.15, 0.90]**	**+1.65 [0.88, 2.42]**	5/5	–

Gains over vanilla on the five-set mean and on ETIS. Sets: test sets, of five, with a positive mean gain. Last column: paired difference on the five-set mean between $a = 2$ and the row.

TABLE IX
COST OF FCT: TRAINING TIME PER EPOCH (MINUTES, MEDIAN OVER SIX SEEDS) AND INFERENCE COST.

Network	Vanilla	FCT	Ratio	Inference
Polyp-PVT	0.56	2.01	3.6×	unchanged
PraNet-V2	0.93	3.11	3.3×	unchanged
ASGNet	1.01	3.70	3.7×	unchanged
Polyp-PVT, no upright-view term	0.56	1.67	3.0×	unchanged

Training time excludes the per-epoch evaluation. One [GPU model] GPU per run. No upright-view term: row (d) of Table VII. Inference: the same network, applied once to the upright image.

and tying them together discourages cues that change under a flip, and such cues are unlikely to transfer to a new centre. Our data support this only indirectly. The flip gap falls on every set, but its fall does not predict where Dice rises, and a single run with flip and rotation augmentation lowered it without any gain.

Scope of the analysis. Proposition 1 needs a binary label and a cross-entropy or squared-error consistency term. It applies to each class of a multi-label task, one class against the rest, but not directly to a softmax over several classes, where the two error vectors need not be parallel and a consistency term can carry information beyond the label. Soft labels also break it, since the two errors may then lie on opposite sides of the target; FCT therefore binarizes the label at 0.5 when it computes the weights, including at rescaled input sizes, where interpolation softens the mask boundary. The analysis also stops at the logits: it does not describe how the network shares the per-pixel terms through its parameters.

Cost. FCT multiplies the training time by about 3.5 and leaves inference unchanged. Without the term on the upright view, the factor drops to three (Table IX) at a similar accuracy (Table VII); applying FCT to part of each batch would lower it further.

Limits of the evidence. The evidence has five limits. First, it covers three networks and one training set of 1,288 images; of the five standard test sets, only ETIS comes from another centre, and on the six PolypGen centres only the confidence-graded variant has been tested so far. Second, the reported epoch is selected on the test sets. The selection is the same for both arms, and the gains remain positive on average without it but are not significant at the last epoch (Fig. 5); the absolute scores are optimistic. Third, the weights were developed over several preliminary variants scored on the same five test sets, so these sets also guided their design. The confidence-graded weight, for instance, was carried to six seeds because of its result on seed 1, which row (h) of Table VII includes; on seeds 2–6 alone, its five-set mean gain is $+0.32$ $[0.08, 0.56]$. The gain of FCT is smallest on seed 1, $+0.18$ points, and is $+0.59$ $[0.15, 1.03]$ on seeds 2–6. Fourth, six seeds resolve gains of about half a point on the five-set mean, but not the smaller differences between variants of the loss, whose CIs are up to several points wide. Fifth, λ was set on Polyp-PVT and not retuned for the other networks, and the study of the loss terms covers Polyp-PVT only, with one row still pending and two at three seeds.

VII. CONCLUSION

We proposed flip-consistent training, a plug-in loss for polyp segmentation that supervises each image in three orientation views and draws their predictions together with a consistency

term. For binary labels, we showed that such a term acts as the supervised loss with an adaptive pixel weight, which is negative where a view is drawn toward a less accurate one, and we gave FCT label-aware weights that remove this case, with a constant fixed by the decision threshold. FCT improved the unseen ETIS set for all three networks we tested, and the five-set mean Dice of Polyp-PVT and PraNet-V2 on every seed, while making their predictions more flip-consistent and leaving inference unchanged. A study of the loss terms attributed most of the ETIS gain to the multi-view supervision; the full loss, with its consistency term and weights, was the only variant that improved all five test sets. Future work will extend the loss to multi-class segmentation and to other transformations, such as rotations, and test FCT itself on the six PolypGen centres, where a label-aware variant already improves on every seed.

Appendix A
Per-Seed Results

Table X lists the mDice of every run behind Tables I and II. FCT scores higher than vanilla with the same seed on the five-set mean for all twelve pairs, and on ETIS for eleven of them.

Appendix B
Gains Without Epoch Selection

Table XI repeats the comparison of Section V-C without epoch selection, with the evaluator that runs during training. At the last epoch, the ETIS gain of Polyp-PVT is the only entry of the table whose CI excludes zero; averaged over the last ten epochs, the gains on the five-set mean and on ETIS are positive for both networks, and their CIs exclude zero except for the five-set mean of Polyp-PVT. For PraNet-V2, the last epoch lies well past the peak of both arms (Fig. 5).

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TABLE X
PER-SEED MDICE (%) OF VANILLA AND FCT ON THE FIVE TEST SETS, BOTH NETWORKS.

	Polyp-PVT: vanilla / FCT						PraNet-V2: vanilla / FCT					
Seed	Kvasir	Clinic	Colon	300	ETIS	Mean	Kvasir	Clinic	Colon	300	ETIS	Mean
1	92.0/92.0	93.3/**93.5**	79.4/**80.4**	88.5/87.0	78.6/**79.8**	86.4/**86.5**	89.0/**89.7**	91.6/90.4	71.8/**72.0**	89.1/**89.7**	65.7/**71.4**	81.4/**82.6**
2	91.0/**92.7**	94.0/93.2	80.8/79.9	89.0/88.3	78.3/**81.3**	86.6/**87.1**	87.9/**89.2**	90.0/89.1	70.1/**73.8**	88.8/**89.5**	69.4/69.2	81.3/**82.2**
3	92.4/**92.8**	92.1/**93.7**	81.0/81.0	87.3/**89.9**	78.7/**80.1**	86.3/**87.5**	85.4/**89.4**	88.6/**90.2**	72.1/**74.4**	82.2/**87.6**	68.8/**69.8**	79.4/**82.3**
4	92.7/92.1	93.5/93.1	79.9/**81.3**	88.9/88.8	78.7/**79.6**	86.7/**87.0**	88.1/**90.7**	90.4/**91.1**	69.7/**72.6**	88.8/87.0	65.3/**73.7**	80.5/**83.0**
5	91.8/**93.0**	93.4/**93.5**	79.7/**79.9**	89.9/89.7	78.3/**80.1**	86.6/**87.2**	87.6/**89.4**	90.2/89.0	70.0/**73.7**	90.4/86.3	67.0/**69.5**	81.0/**81.6**
6	91.6/**92.4**	92.7/**93.1**	80.8/80.1	89.1/**89.4**	79.1/**80.8**	86.7/**87.2**	89.7/89.4	88.0/**89.2**	65.7/**71.0**	87.0/86.6	66.2/**70.5**	79.3/**81.3**

Bold: FCT above vanilla with the same seed (before rounding). Clinic: CVC-ClinicDB; Colon: CVC-ColonDB; 300: CVC-300.

TABLE XI
GAINS OF FCT WITHOUT EPOCH SELECTION: THE TRAINING-TIME EVALUATOR AT THE LAST EPOCH AND AVERAGED OVER THE LAST TEN EPOCHS (POINTS, 95% CI, SEEDS BETTER). BOLD: CIS THAT EXCLUDE ZERO.

Epochs	Five-set mean	ETIS
Polyp-PVT		
Last epoch	+0.55 [−0.11, 1.22], 4/6	**+2.60 [0.84, 4.36], 6/6**
Last 10 epochs	+0.42 [−0.03, 0.88], 5/6	**+1.52 [0.11, 2.93], 4/6**
PraNet-V2		
Last epoch	+0.49 [−1.17, 2.14], 4/6	−0.18 [−2.20, 1.84], 2/6
Last 10 epochs	**+2.41 [1.40, 3.43], 6/6**	**+3.71 [0.75, 6.68], 5/6**
ASGNet		
Last epoch	+0.13 [−0.91, 1.17], 3/6	+1.69 [−1.01, 4.39], 4/6
Last 10 epochs	+0.21 [−0.59, 1.01], 4/6	+1.99 [−0.26, 4.24], 4/6

Training-time evaluator: Dice of the min-max normalized probability map, per image, as in the official Polyp-PVT code. The six-seed mean of FCT is above that of vanilla after 81 of 99 epochs for Polyp-PVT, 28 of 29 epochs for PraNet-V2, 76 of 79 epochs for ASGNet.

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